THIEN TRANDINH, MASc

Software Engineer

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SKILLS

Languages & Frameworks: **TypeScript, C#, .NET, SQL, JavaScript, React, Unity**
AI-Augmented Development: **AI Coding Agents (GitHub Copilot, Codex), Prompt Engineering**
Systems & Architecture: **RESTful APIs, Distributed & Scalable Systems, Microservices, Entity Component System (ECS)**
Tooling & Delivery**: Git, CI/CD, Jira, Confluence, Growthbook (A/B Testing)**
Engineering Practices: **SDLC, OOP, Design Patterns, Code Reviews, Unit Testing, Agile/Scrum**

JOB EXPERIENCE

Prodigy Education, *Software Engineer* • *C#, TypeScript • Unity* — *Feb 2025 – Jan 2026*

- Main contributor to **Prodigy Math'**s Replay Campaign, a scalable post-completion progression system that increased **subscription conversion by 8.6% and monthly revenue by 11%**
- Built modular ECS components following SOLID principles and performed cost analysis on legacy code, reducing **content iteration time by 50%+** and lowering long-term maintenance costs
- Gathered requirements, designed and delivered the Linear Campaign onboarding system, **reducing new-player churn by 15% and improving subscription conversion by 4%** through A/B testing and staged rollouts
- Served as a **core engineer on Prodigy's rapid-prototyping Game Island initiative** delivering monetized and performance optimized experiences at production scale across web and Chromebook platforms
- Led development of Neek's Bubble Rescue and Academy Defense, integrating education, progression, and monetization frameworks contributing to **increasing player retention by 8%**
- Built internal designer & QA tooling including Spreadsheet-to-JSON **pipelines that reduced data iteration time by 90%** and debug commands and tooling that significantly improved QA throughput and regression testing
- Applied **AI coding agents with strong engineering judgment to accelerate development and decision-making** across features and systems while upholding production-grade quality standards
- Owned multiple live, user-facing features end-to-end, including onboarding flows, interactive content modules, and tutorial systems, and acted as **release owner responsible for production stability and hotfixes**

Gatarn Games Ltd, *Full-Stack Software Engineer* • *C# .NET, TypeScript, REST, SQL* — *Dec 2022 – Dec 2024*

- **Led development of Skillfite.io**, a web-based real-time application with **1,000+ daily active users**
- Designed and implemented **monetization systems** including one-time purchases, banner ads, and video ads supporting revenue growth
- Designed and implemented backend systems powering player progression, monetization, and real-time user interactions
- Built **authentication and session services** connecting web clients to backend services via TCP, WebSockets, and Google OAuth tokens
- Implemented a **SQL-backed player data platform** to support accounts, inventory, and progression
- Architected and optimized scalable, high-availability runtime systems**, reducing client–server network traffic by ~90%, improving latency and lowering server costs**
- Designed a reusable **ECS architecture** enabling modular, data-driven feature development and faster iteration.
- Implemented an **object pooling framework** that improved CPU and memory performance by **80%**

- Built the TypeScript web client powering gameplay, UI, and monetization, including **reusable UI frameworks and SDK-agnostic ad-integration layers** supporting multiple providers
- Drove performance and reliability improvements across the live service using runtime profiling**, improving overall performance by ~50%**
- **Integrated AI-assisted development workflows** to safely accelerate large-scale refactoring, test generation, and system changes in a live production environment
- Led engineering delivery and team operations, running standups, sprint planning, reviews, and retrospectives using Jira
- Designed and ran the technical hiring process, interviewing and onboarding new engineers

Imagine Communications, *Software Engineer* • *TypeScript, C# .NET, VUE* — *Nov 2021 – Dec 2022*

- Designed and maintained **containerized web applications** using Vue.js, TypeScript, and Docker
- Worked within a SAFe Agile environment, collaborating across teams and **contributing to CI/CD pipelines** using Git, Bitbucket, Jenkins, and Jira
- Improved customer usability through dynamic landing pages and custom table filtering
- Improved code maintainability by taking an active role in bi-weekly merges of long-term branches

Viral Staging, *Freelancer Developer* • *JavaScript* — *May 2022 – Aug 2022*

- Implemented an embeddable HTML/JavaScript geolocation widget, resolving user postal codes to the nearest of **700+ vendors**
- Built **Google Maps + Google Drive API integrations** to dynamically display location data on customer-facing websites

McMaster Children's Hospital, *Software Engineer* • C#, Unity — *2019 – 2021, 2026 – present (Contract)*

- Designed, developed, and maintained a VR-based rehabilitation application currently used in hospital clinics
- Owned the system **end-to-end,** from initial prototyping and research through deployment and ongoing maintenance in a clinical environment
- Designed systems with safety, correctness, and usability constraints appropriate for patient-facing clinical software
- Direct stakeholder collaboration (doctors and physiotherapists) to translate clinical requirements and **safety constraints into technical system designs**
- Leveraged **Google Maps API to dynamically obtain terrain data,** and designed tuning methods to transform terrain data into a traversable game map
- **Iteratively improved the application** based on clinician feedback, patient usage data, and observed therapy outcomes

Evertz, *Software Design Engineer Internship* • JavaScript — *May 2017 – Aug 2018*

- Developed and maintained UI control applications, and implemented **third-party hardware integration**
- Built a Unity-based simulation controllable via TCP commands to support testing and visualization of live sports production systems

EDUCATION & CERTIFICATIONS

McMaster University, Hamilton, Ontario

Software Engineering MASc. (Master of Applied Science) (Link to thesis) — *2019 –2022*
Bachelor of Software Engineering — *2014 – 2019*

Certifications: *Microsoft Azure Fundamentals (AZ-900)* — *2024*